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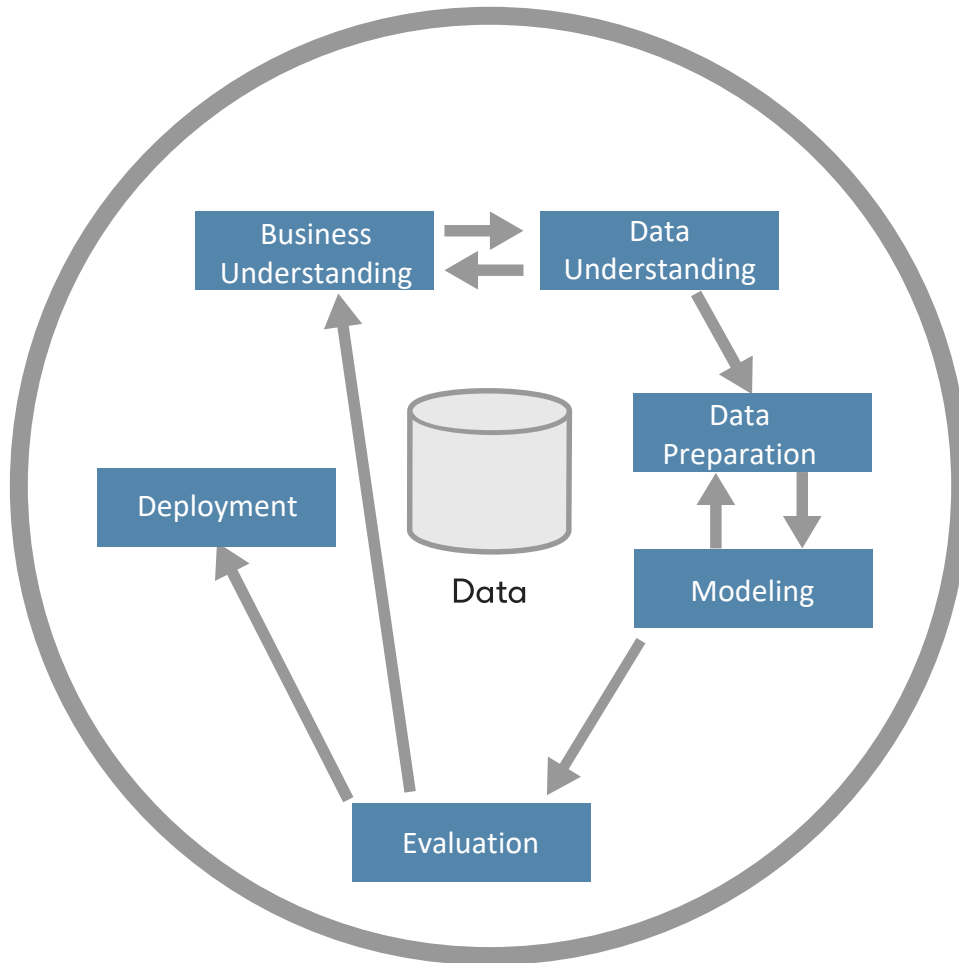
Research Unit of Production and Maintenance
Management

Industrial Data Science

Exercise: Exploring Machine Learning Techniques Using Dry Room Data

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Optional Exercise: TU Braunschweig Collaboration [1 ECTS]



Business & Data Understanding: Get insights on the topic of battery cell production to identify the main challenges and needs within the process. Discuss why data analysis is required and what potential implications it may have for improving production.

Date: 11.05.2026, 13:00-14:00 (**Introduction:** Dr.-Ing. Gabriela Ventura Silva;
<https://tuwel.tuwien.ac.at/mod/url/view.php?id=2971406>)

Data Preparation & Modeling: Based on the supervised and unsupervised approaches covered in the course, identify relevant information. Deploy and apply suitable machine learning models. (TU Wien)

Evaluation: Evaluation of the developed models using the metrics provided by the TU Braunschweig colleagues. (TU Wien)

Deployment: Based on the findings and the exchange activities between TU Wien and TU Braunschweig on the scheduled dates, derive the implications of your models by focusing on minimizing total costs and CO₂ emissions, and develop a structured work plan accordingly.

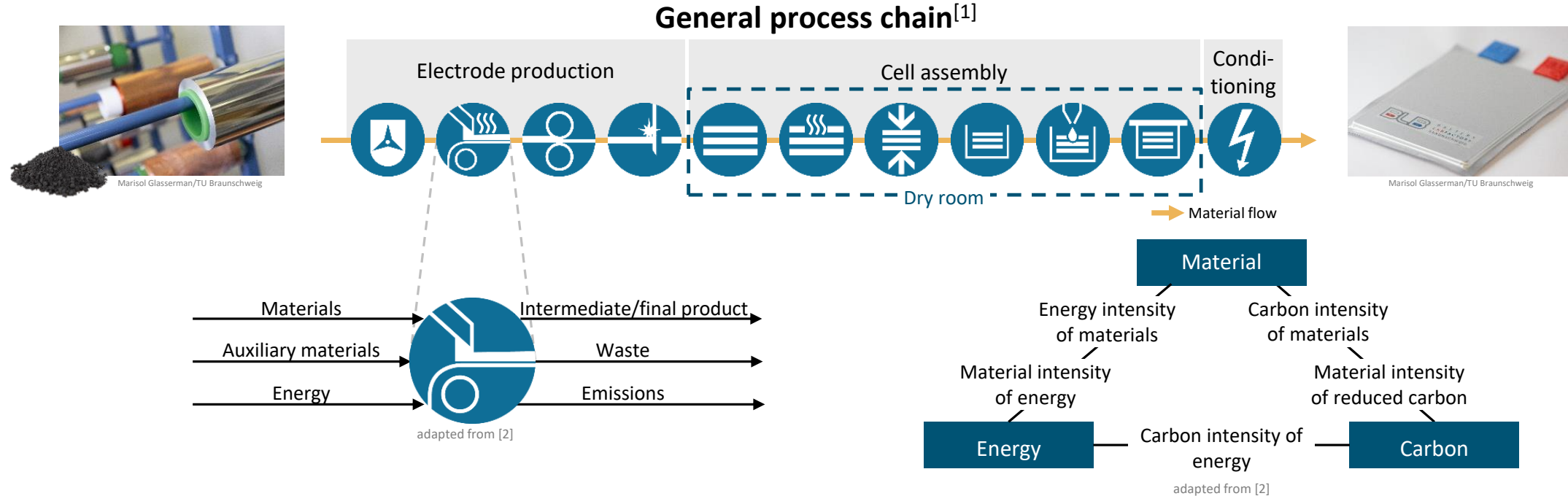
Presentation: Present your results and insights at the joint colloquium.

Date: 16.06.2026, 15:45-17:00 (**10-15min per group incl. discussion; Webex**)
Limited number of slots available (up to 20 participants)

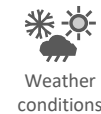
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Business Understanding

Battery cell production



- Foreground system: **Individual elements** coupled by **flows**
- Background system: **Influencing** the production system



[1] Kwade et al. (2018): "Current status and challenges for automotive battery production technologies"

[2] Hauschild et al. (2018): "Life cycle assessment: Theory and practice"

Business Understanding

Dry room in battery cell production

Importance of dry rooms

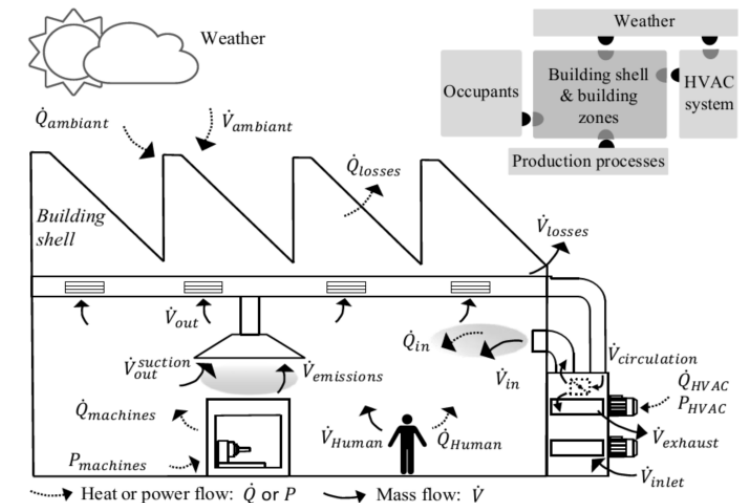
- Excessive moisture during production affects the electrochemical performance of battery cells, therefore critical processes take place inside of dry rooms
- Dry room is a controlled environment with temperature around 20 °C and low relative humidity (dew point between -40 and -60 °C)¹
- Significant contributor to high energy demand, accounting for up to 60 % of total energy use in pilot-scale production²

Influencing factors of dry room operation and energy demand

- Internal: Personnel and the associated heat and moisture, heat loss from machines
- External: Outside temperature, relative humidity

Data source

- Physics-based model of dry room energy demand for a battery cell pilot line³
- Main modeled system components: fan, drying unit, heat exchanger, and the drying room itself



Schematic representation of involved system elements

¹ Michaelis et al. (2023). *Roadmap Battery Production Equipment 2030. Update 2023*. VDMA Verlag.

² Thomitzek et al. (2019). "Simulation-based assessment of the energy demand in battery cell manufacturing". In: *Procedia CIRP* 80, pp. 126–131.

³ Vogt, M. (2023). "Planning and operation of HVAC systems in production environments". Dissertation. Vulkan.

Data Understanding

In total, **four datasets** are available:

1. *dry room 1_input* (from a physic-based simulation model)
2. *dry room 1_output* (from a physic-based simulation model)
3. *DE_2023_hourly*
4. *Electricity_production_and_spot_prices_in_Germany_in_2023*

Data Understanding

Dry room input dataset includes:

- Internal variables: heat gains from machinery (*heatload*), moisture and heat contributions from personnel (*numOfHum*)
- External variables: ambient temperature (*outsideTemp*), ambient relative humidity (*relHumidity*)

time	numOfHum	outsideTemp	relHumidity	heatload
01.01.2023 00:00	2	26.32004527	81.37606438	0
01.01.2023 01:00	1	25.60261767	82.72641587	0
01.01.2023 02:00	2	24.82616063	82.85339315	0
01.01.2023 03:00	2	25.30131476	83.83867368	0
01.01.2023 04:00	0	24.81392743	83.89842714	0
01.01.2023 05:00	1	25.34044693	84.59053094	1768.819
01.01.2023 06:00	1	24.20798198	84.25020323	574.266
01.01.2023 07:00	2	25.84309038	81.27488838	2000.089
01.01.2023 08:00	1	27.23259499	75.12014423	2046.343
01.01.2023 09:00	2	27.34823052	71.69774347	1123.644
01.01.2023 10:00	2	29.67667207	70.78826597	1825.767
01.01.2023 11:00	2	29.27094528	69.68008283	3444.098
01.01.2023 12:00	0	30.34763453	69.5537178	3582.86

Data Understanding

Dry room output dataset includes:

The output dataset reflects the energy demand of the dry room, which is the main variable of interest. These outputs are directly influenced by the input variables and represent the energy required to maintain strict environmental conditions necessary for battery cell production.

- Electricity demand (*electricity_W*)
- Heat demand (*heat_W*)

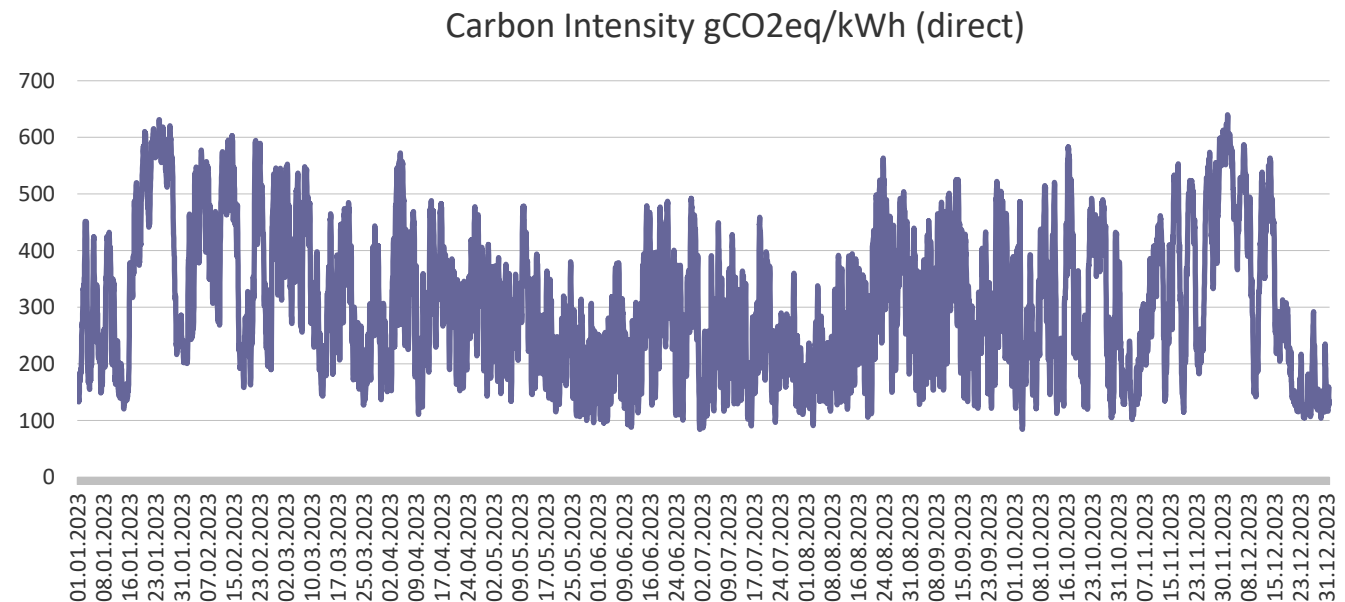
timestamp	electricity_W	heat_W
01.01.2023 00:00	46700.04464	68184.3151
01.01.2023 01:00	45819.56008	68446.1577
01.01.2023 02:00	44904.43172	69317.7068
01.01.2023 03:00	45459.62909	68943.5627
01.01.2023 04:00	44890.61253	68794.4653
01.01.2023 05:00	45506.33769	67839.5186
01.01.2023 06:00	44209.87665	69339.2555
01.01.2023 07:00	46106.85979	67528.9421
01.01.2023 08:00	47874.8096	66255.0163
01.01.2023 09:00	48030.13878	67051.9523

Data Understanding

Carbon Intensity Data-(DE_2023_hourly.csv)

This data describes how energy was produced at each point in time. It shows that emissions vary over time depending on the electricity mix, with lower emissions typically occurring during periods of higher renewable generation.

Datetime (UTC)	Carbon Intensity gCO2eq/kWh (direct)	Renewable Percentage
01.01.2023 00:00	140.51	78.42
01.01.2023 01:00	140.44	78.93
01.01.2023 02:00	148.34	77.74
01.01.2023 03:00	149.55	77.51
01.01.2023 04:00	148.04	77.74
01.01.2023 05:00	148.79	77.55
01.01.2023 06:00	146.65	77.95



- Source: <https://app.electricitymaps.com>

Data Understanding

Electricity Generation and Price Data(*Electricity_production_and_spot_prices_in_Germany_in_2023.csv*)

This dataset provides information about the hourly electricity generation by source and corresponding market prices. It includes power generation from:

- Nuclear
- Non-Renewable sources
- Renewables source
- Additionally: a variable describing the day-ahead electricity price.

Date (GMT+1)	Nuclear Power (MW)	Non-Renewable Power (MW)	Renewable Power (MW)	Day Ahead Auction (DE-LU) Price (EUR/MWh, EUR/tCO2)
2023-01-01T00:00+01:00	2459.17	9265.614572	36641.97777	-5.17
2023-01-01T01:00+01:00	2458.6025	9042.471891	37696.83394	-1.07
2023-01-01T02:00+01:00	2459.645	8955.260549	37849.30209	-1.47
2023-01-01T03:00+01:00	2460.475	8967.997791	35565.44802	-5.08
2023-01-01T04:00+01:00	2460.8025	8971.925269	35084.99508	-4.49
2023-01-01T05:00+01:00	2465.0175	8920.641783	35360.22811	-5.4
2023-01-01T06:00+01:00	2464.4575	8921.075666	35289.54158	-5.02
2023-01-01T07:00+01:00	2464.2175	8957.341973	35994.39361	-1.3

- Source: https://www.energy-charts.info/charts/price_spot_market/chart.htm?l=en&c=DE&year=2023&interval=year&minuteInterval=60min&legendItems=3x0tg0&stacking=sparkline

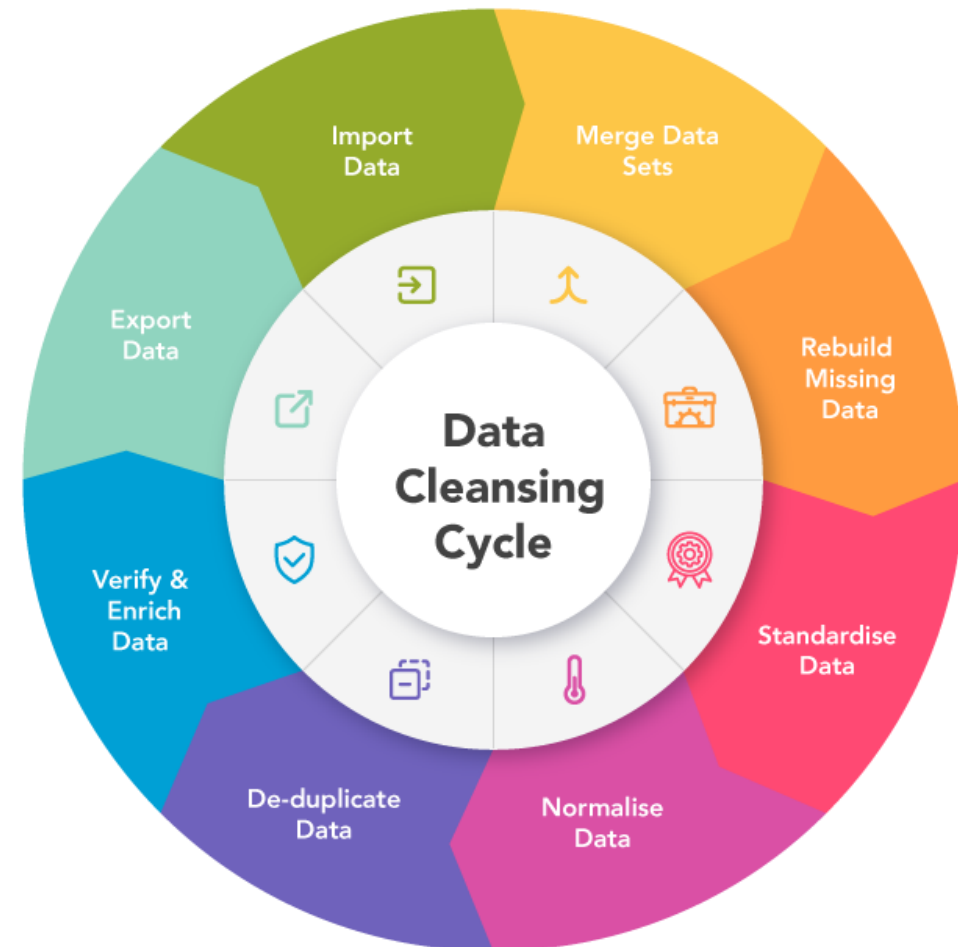
Data Preparation

Task 1:

Prepare the available datasets following the data cleansing cycle.

Deliverable:

A merged dataset (all 4 files) containing all variables required to estimate electricity and heating demand. It should also include the variables needed to analyze optimal working time windows, specifically identifying green and low-cost periods.



• Source Image: <https://www.melissa.com/>

Modeling

Task 2:

Exploratory Data Analysis (EDA):

- Correlation Analysis: Build a correlation matrix to identify relationship between input variables and output variables.
- Identification main drivers: identify the main drivers of energy consumption by calculating correlation strengths and comparing the relative influence of internal and external factors.

Deliverable: Correlation matrix

Task 3:

Develop predictive models for estimating electricity and heat demand based on input variables. Machine learning algorithms can be selected and justified based on their applicability to the problem.

- Unsupervised Learning (e.g. Clustering)
- Supervised Learning (e.g. Decision tree)

Deliverable: 1 model needs to be selected and implemented.

- Source

Evaluation & Deployment

- Evaluate the performance of electricity and heat demand forecasting models.
- Investigate whether identifiable optimal operating time windows exist for the dry room, specifically periods characterized by low electricity cost, low system demand, and high renewable energy availability (“green and cost-efficient periods”).

Task 4: Based on the findings try to answer following questions for a more low-cost and low-carbon.

- At which time periods is electricity both **low-cost and low-carbon**?
- Are there identifiable **optimal working time windows** (i.e., “green and cost-efficient periods”) for operating the dry room, based on the available datasets? Assumptions, like working hours act, shift operations, weekends or public holidays can be added.

- Source

Task 4 Outcome Template

Period:	Recommendation Level:	Rationale
– Season: "spring" – Months: ["04", "05"] – Days: ["Monday", "Tuesday"] – primary_hours: - "08:00–15:00" – secondary_hours: - "15:00–21:00"	"high"	E.g. These months provide the best overlap between lower electricity prices and lower carbon intensity, especially during daytime and early evening hours when renewable generation is typically stronger.

Deliverables

Deliverables:

- **Presentation [10min per group + 5 min Q&A; 5-10 slides; 50% of the grade; submitted via TUWEL]:**
E.g., Summary of methodology, findings, and key insights.
- **Jupyter Notebook [30% of the grade; upload to JupyterHub]:**
Reproducible workflow including data preprocessing, exploratory analysis, model evaluation, and time-window identification.
- **Working Time Window Summary [20% of the grade; submitted via TUWEL]:**
A ranked table of the top three recommended working time windows for operating the dry room, based on combined conditions of low electricity prices, low demand, and high renewable energy generation.



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